Reversing a conditional probability

High school · probability · Progressive exercises · 07

Given information. For exercises 1 and 2: choose urn A with probability $\frac{1}{3}$ and urn B with probability $\frac{2}{3}$. A contains 3 red balls and 1 blue; B contains 1 red and 3 blue. Then draw a ball uniformly from the chosen urn.

Exercise 1. In the example, find $P(A \cap \text{blue})$.

Exercise 2. Find $P(A|\text{blue})$.

Exercise 3. In a class, 12 of 30 students play chess; 9 of those enjoy puzzles. Find $P(\text{puzzles}|\text{chess})$. Can $P(\text{chess}|\text{puzzles})$ be deduced?

Exercise 4. Events A and R satisfy $P(A) = 0.4$, $P(R|A) = 0.5$ and $P(R|\text{not } A) = 0.5$. Find $P(A|R)$.